

BEGINNING OF TRANSMISSION

State: unpublished / pending pairing

0.0.1 Release metadata

- **Title:** Active Inference Multi-Track Exemplar
- **Version:** 0.1.0
- **DOI:** pending
- **GitHub:** docxology/template_active_inference
- **SHA-256:** fddea89a41b87eaf9e2e88a0745dd6cb28ff1138f5cd3b81857d716b9d0f219e
- **SHA-512:** b0dc133c69ad5a5eb6ec54d9701d123c21438a2ec212fcba66cc4aac605ddf217ec2c6d84ab49e5a26e47795e307bc9d784affcd3a5d3b039f420a18fdb65c87

Pairing: pending — unresolved: - DOI minted: pending - GitHub release URL: pending

Figure 1: Integrity QR strip

0.0.2 Transmission manifest

```
title=Active Inference Multi-Track Exemplar
version=0.1.0 doi=pending
sha256=fddea89a41b87eaf... manifest={"t":"Active Inference Multi-T","v":"0.1.0","d":"","s":"fddea89a41b87eaf"}
Structured manifest: ../data/transmission_manifest.json
```

Figure 2: Publication pairing flow

Stego: off | overlays text | barcodes on | XMP on | manifest on → ./secure_run.sh

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1 Sheaf Track Coverage

This page summarizes which **sheaf fragment tracks** are bound for each IMRAD row in `manuscript/sheaf/manifest.yaml`. The matrix is regenerated at compose time.

Totals: 42 present / 42 bound / 0 missing (gray).

Color	Meaning
Black	Track present (bound and fragment exists)
White	Absent (not bound for this row)
Gray	Missing (bound but fragment file absent)

1.1 Introduction

- **Introduction** (*group*)
- **Motivation and Active Inference scope**
- **Exemplar contributions** ## Methods
- **Methods** (*group*)
- **Analytical Bernoulli–Ising toy**
- **pymdp sophisticated inference harness**
- **Lean boundary witnesses**
- **Sheaf composition pipeline** ## Results
- **Results** (*group*)
- **Mutual information sweep**
- **Free energy decomposition**
- **T-maze sophisticated inference rollout**
- **Invariant gate summary** ## Discussion
- **Discussion** (*group*)
- **Limitations and extensions** ## Appendix
- **Full sheaf track coverage (proof)**

Figure 4. Sheaf track coverage for 16 manifest rows and 10 fragment tracks (42 present / 42 bound / 0 missing).

Appendix row `16_appendix_full_sheaf.md` binds every registered track type as a composability proof.

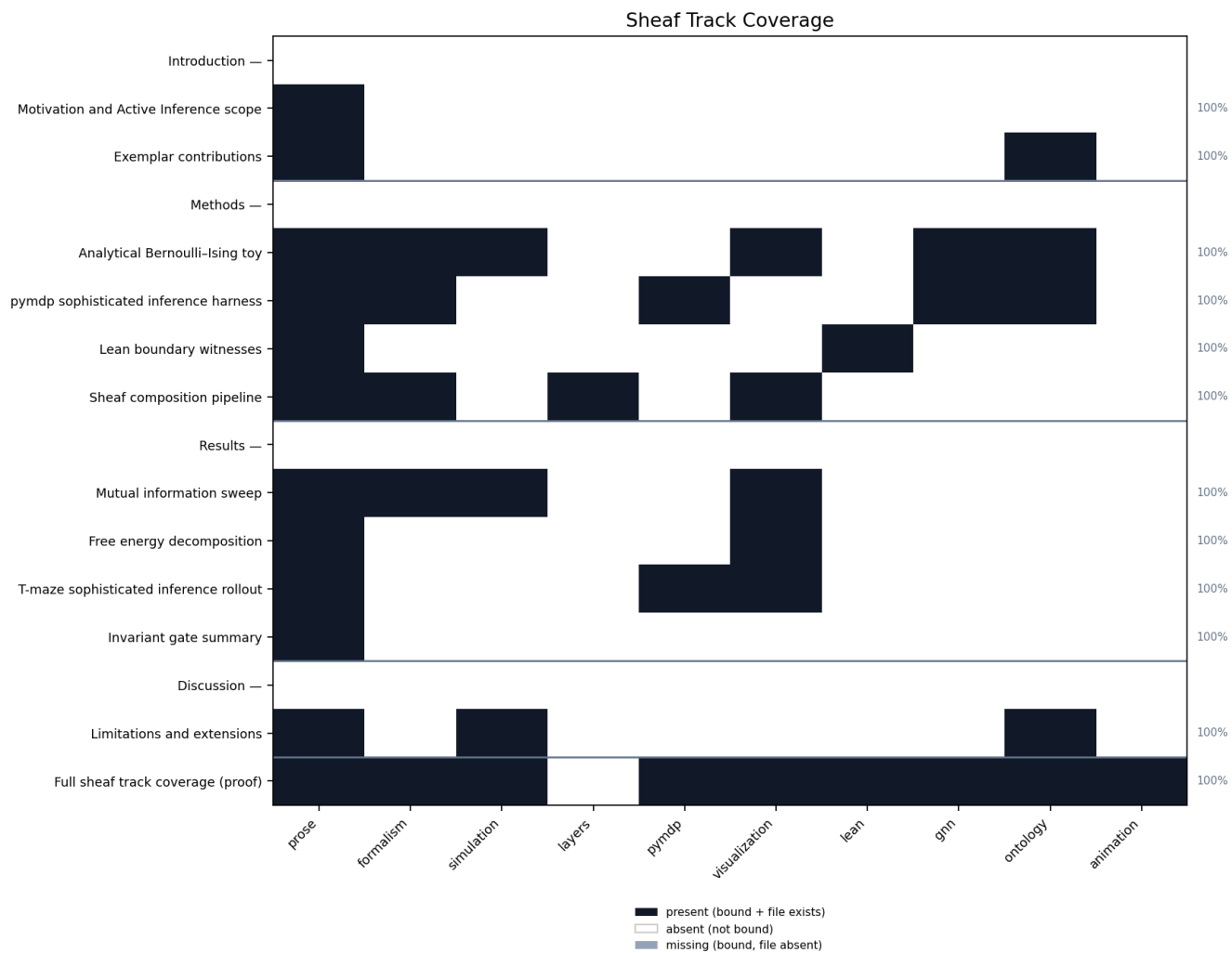


Figure 3: Heatmap matrix of IMRAD manuscript rows versus nine sheaf fragment track columns. Black cells mean the track is bound and the fragment file exists; white cells mean the track is not bound; gray cells mean bound but missing. Rows are grouped by IMRAD block with indented subsection labels; column headers list track ids.

2 Abstract

This public exemplar binds **7 pipeline tracks** — Lean, analytical Python, pymdp sophisticated inference, GNN, ontology concordance, visualizations, and the sheaf manuscript composer — declared in `tracks.yaml`. Flat manuscript sections follow an **IMRAD outline** (Introduction, Methods, Results, Discussion, plus appendix) assembled from **10 sheaf fragment types** registered in `manuscript/sheaf/tracks.yaml`.

The first page (`00_00_sheaf_coverage.md`) shows a **16-row coverage matrix** (4 IMRAD group headers and 12 composed subsections, including a full-track appendix proof) regenerated from the live manifest at compose time.

The T-maze demo aligns with `pymdp sophisticated_inference examples`.

Measured invariant checks: 6 / 6 passed. SI planning horizon: 2 steps.

3 Motivation and Active Inference scope

This public exemplar demonstrates a **sheaf-composed** Active Inference manuscript: flat sections bind optional fragment tracks (prose, formalism, simulation, pymdp, visualization, Lean, GNN, ontology) under an IMRAD outline. The first page heatmap summarizes which tracks are bound per section.

The pymdp track follows the **pymdp sophisticated_inference examples** with a minimal T-maze and planning horizon `policy_len = 2`.

4 Exemplar contributions

This exemplar contributes:

1. **7 pipeline tracks** (`tracks.yaml`) — analytical Python, pymdp, GNN, ontology, Lean, visualizations, and sheaf composition — each with artifact gates.
2. **10 sheaf fragment types** (`manuscript/sheaf/tracks.yaml`) — composable prose, formalism, simulation, pymdp, visualization, Lean, GNN, ontology, and optional animation.
3. A **16-row IMRAD coverage matrix** on the first page, regenerated from the live manifest at compose time.
4. Registry-backed **invariant checks** (6 / 6 passed) and measured SI rollout metrics.

4.0.1 Ontology bindings

- `expected_free_energy` → `ExpectedFreeEnergy`
- `location` → `HiddenState`
- `observation` → `ObservationLikelihood`
- `policy` → `PolicyPosterior`

5 Analytical Bernoulli–Ising toy

We study a minimal **K=2 Bernoulli / Ising** coupling as the analytical companion to multi-track verification. Closed-form mutual information $I(\lambda)$ provides an oracle for empirical checks and GNN round-trips.

Measured sweep grid points: 21. Invariants passed: 6 / 6.

The entangled joint over binary policies satisfies

$$q_\lambda(\pi) \propto E(\pi) \exp(\lambda J(\pi)),$$

with symmetric Ising coupling J and deformation parameter λ . Mutual information is $I(\lambda) = \log 2 - H_b(\sigma(\lambda))$.

The analytical track writes `output/data/parameter_sweep.csv` comparing closed-form and empirical mutual information across $\lambda \in [0, 4]$.

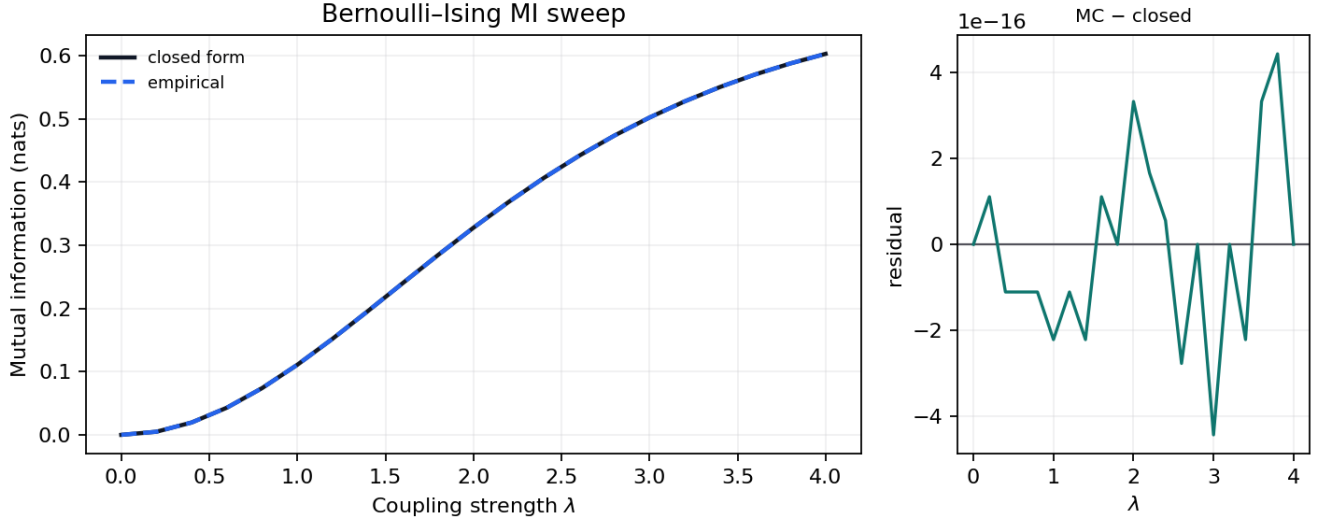


Figure 4: Two-panel plot of mutual information versus coupling strength λ for the symmetric Bernoulli-Ising toy. Left panel shows closed-form curve (solid dark line) and Monte Carlo estimates (dashed blue line) with λ on the x-axis and mutual information in nats on the y-axis. Right panel shows Monte Carlo minus closed-form residuals versus λ with a zero reference line.

Figure 1 (methods). Closed-form and Monte Carlo estimates of pairwise mutual information in the symmetric Bernoulli-Ising toy; saturation approaches $\log 2$ nats.

The Bernoulli toy is also declared in `gnn/bernoulli_toy.gnn.md` (GNN v1.1). Ontology annotations bind framework symbols to Active Inference Ontology terms.

5.0.1 Ontology bindings

- $J \rightarrow$ **CrossStreamCouplingPotential**
- $\gamma \rightarrow$ **SophisticationWeight**
- $\lambda \rightarrow$ **EntanglementDeformationParameter**
- $\pi_1 \rightarrow$ **Stream1PolicyVector**
- $\pi_2 \rightarrow$ **Stream2PolicyVector**
- $q_{\text{joint}} \rightarrow$ **EntangledJointPosterior**

6 pymdp sophisticated inference harness

Sophisticated inference (planning horizon). This section documents a **real pymdp state-inference harness** on a minimal T-maze with planning horizon `policy_len = 2`. The Agent constructs a multi-step policy set (`num_policies` logged in artifacts); per-step **belief entropy** is recorded in `output/logs/pymdp_runs.jsonl` and aggregated as `mean_belief_entropy`.

Full policy-posterior `infer_policies` on this toy is an opt-in extension track — see `tracks.yaml` `graph_world`. Reference notebooks: [pymdp sophisticated_inference](#).

Mean belief entropy across steps: 0.3251.

Given generative matrices A, B, C, D , pymdp computes state beliefs $q(s)$ via variational inference (`infer_states`). The Agent is configured with planning horizon $H = 2$, which defines the **policy depth** used when constructing candidate policies (logged as `num_policies` in `output/data/si_tmaze_summary.json`).

This exemplar records belief entropy per step; extending to full expected-free-energy policy selection (`infer_policies`) is documented as a follow-on track.

Artifact paths:

- `output/logs/pymdp_runs.jsonl` — append-only JSONL run log
- `output/data/si_tmaze_summary.json` — step count, actions, mean belief entropy
- `output/data/si_tmaze_trace.json` — rollout trace

Steps recorded: 2.

See `gnn/si_tmaze.gnn.md` for a GNN view of the T-maze hidden state, observation, and policy variables with ontology bindings.

6.0.1 Ontology bindings

- `belief_entropy` → **BeliefEntropy**
- `loc` → **HiddenState**
- `obs` → **ObservationLikelihood**
- `pi` → **PolicyPosterior**

7 Lean boundary witnesses

The Lean track provides minimal boundary witnesses checked by `lake build` under `lean/TemplateActiveInference/`. Fragments cite theorem names without duplicating proof scripts in prose.

See the Lean fragment for module references aligned with the analytical toy.

Lean module `TemplateActiveInference.SophisticatedInference` declares the planning horizon parameter and a witness that $\text{horizon} > 1$ distinguishes sophisticated from myopic inference.

Build via `lake build` under `lean/`.

8 Sheaf composition pipeline

Sheaf composition glues registered fragment tracks into flat manuscript sections using `manuscript/sheaf/manifest.yaml` and `manuscript/sheaf/tracks.yaml`.

Figure 6 summarizes the 10 fragment layer types and their IMRAD bindings in one overview (registry stack plus binding heatmap). The tables below list every track definition and section×track binding from the live manifest at compose time.

```
uv run python scripts/compose_manuscript.py
```

```
uv run python scripts/compose_manuscript.py --validate-only --strict
```

Each compose run emits `output/data/sheaf_coverage_matrix.json` and regenerates coverage artifacts. Partial compose (`--section`) is draft-only; the matrix always reflects the full manifest.

Coverage semantics: **black** / **P** = present (bound and file exists); **white** / **—** = absent (not bound); **gray** / **M** = missing (bound but file absent). Discussion limitations reference the same matrix counts (42 / 42 / 0).

Formal definitions for the composable fragment track types appear in the registry table below. Bindings for each IMRAD manifest row are listed in the binding matrix table.

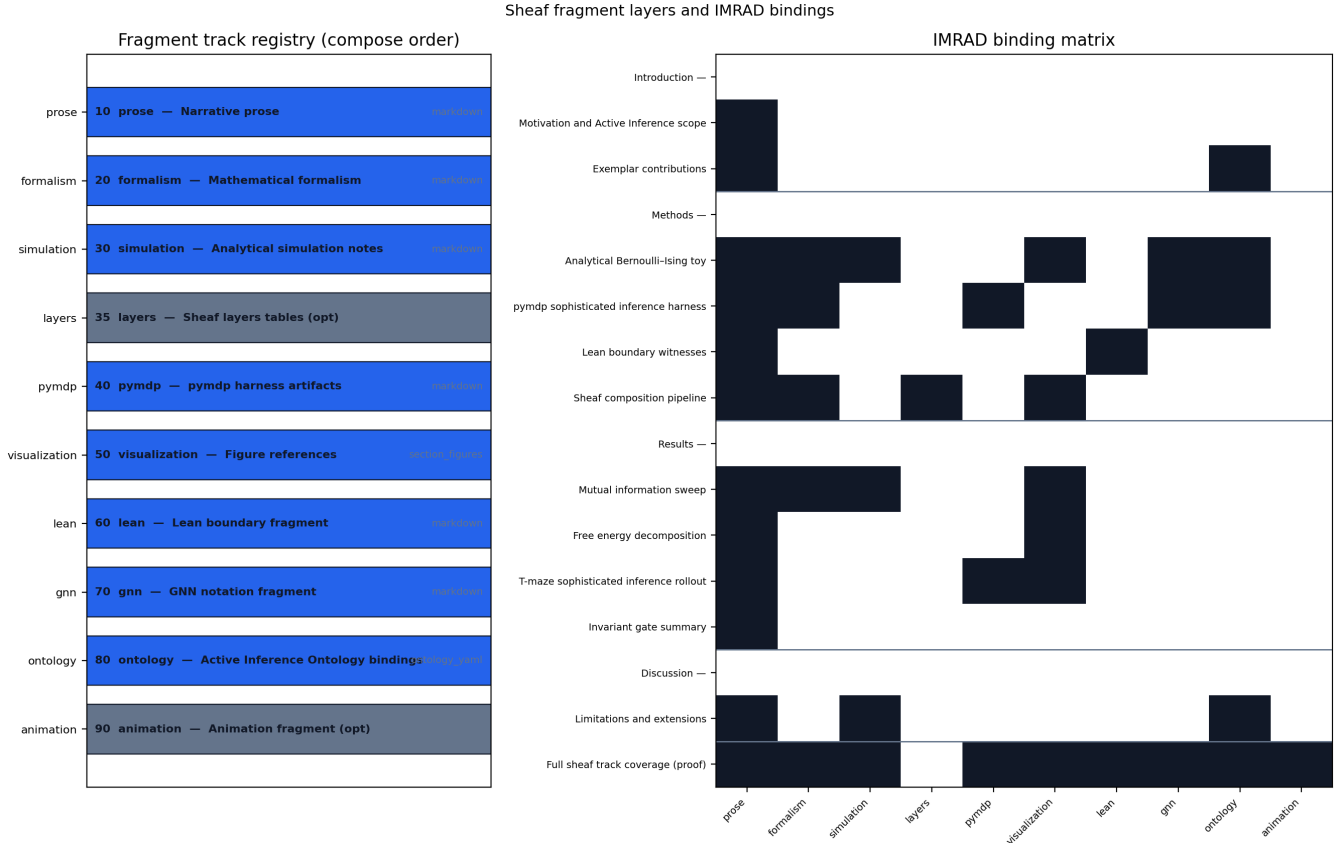


Figure 5: Two-panel overview of sheaf fragment layers. Left panel shows nine composable track types in registry compose order with labels and renderer ids. Right panel shows the IMRAD section binding heatmap with black present, white absent, and gray missing cells across 16 manifest rows and 10 tracks.

Figure 6 (methods). Sheaf fragment track registry (left) and IMRAD binding matrix (right) for 10 layer types (42 present / 42 bound / 0 missing).

8.1 Sheaf fragment track registry

Compose order and renderer bindings from `manuscript/sheaf/tracks.yaml`.

	Order	Track id	Label	Renderer	Optional
	10	<code>prose</code>	Narrative prose	<code>markdown</code>	No
	20	<code>formalism</code>	Mathematical formalism	<code>markdown</code>	No
	30	<code>simulation</code>	Analytical simulation notes	<code>markdown</code>	No
	35	<code>layers</code>	Sheaf layers tables	<code>layers_report</code>	Yes
	40	<code>pymdp</code>	pymdp harness artifacts	<code>markdown</code>	No
	50	<code>visualization</code>	Figure references	<code>section_figures</code>	No
	60	<code>lean</code>	Lean boundary fragment	<code>markdown</code>	No
	70	<code>gnn</code>	GNN notation fragment	<code>markdown</code>	No
	80	<code>ontology</code>	Active Inference Ontology bindings	<code>ontology_yaml</code>	No
	90	<code>animation</code>	Animation fragment	<code>markdown</code>	Yes

Track count: 10 registered fragment types.

8.2 IMRAD binding matrix

Section rows versus fragment track columns. **P** = present (bound and file exists); **—** = absent (not bound); **M** = missing (bound, file absent).

Section	<code>prose</code>	<code>formalism</code>	<code>simulation</code>	<code>layers</code>	<code>pymdp</code>	<code>visualization</code>	<code>lean</code>	<code>gnn</code>	<code>ontology</code>	<code>animation</code>
Introduction (group)	—	—	—	—	—	—	—	—	—	—
Motivation and Active Inference scope	P	—	—	—	—	—	—	—	—	—
Exemplar contributions	P	—	—	—	—	—	—	—	P	—
Methods (group)	—	—	—	—	—	—	—	—	—	—
Analytical Bernoulli-Ising toy	P	P	P	—	—	P	—	P	P	—
pymdp sophisticated inference harness	P	P	—	—	P	—	—	P	P	—
Lean boundary witnesses	P	—	—	—	—	—	P	—	—	—

Section	prose	formalism	simulation	layers	pymdp	visualization	mean	gnn	ontology	animation
Sheaf composition pipeline	P	P	—	P	—	P	—	—	—	—
Results (group)	—	—	—	—	—	—	—	—	—	—
Mutual information sweep	P	P	P	—	—	P	—	—	—	—
Free energy decomposition	P	—	—	—	—	P	—	—	—	—
T-maze sophisticated inference rollout	P	—	—	—	P	P	—	—	—	—
Invariant gate summary	P	—	—	—	—	—	—	—	—	—
Discussion (group)	—	—	—	—	—	—	—	—	—	—
Limitations and extensions	P	—	P	—	—	—	—	—	P	—
Full sheaf track coverage (proof)	P	P	P	—	P	P	P	P	P	P

Totals: 42 present / 42 bound / 0 missing.

Symbol	Coverage color	Meaning
P	Black	Track present (bound and fragment exists)
—	White	Absent (not bound for this section)
M	Gray	Missing (bound but fragment file absent)

9 Mutual information sweep

We sweep coupling strength λ on a grid of 21 points up to $\lambda_{\max} = 4$. Closed-form mutual information $I(\lambda)$ is compared to Monte Carlo estimates from the analytical module.

Measured invariant checks: 6 / 6 passed on the clean tree.

The entangled joint over binary policies satisfies

$$q_{\lambda}(\pi) \propto E(\pi) \exp(\lambda J(\pi)),$$

with symmetric Ising coupling J and deformation parameter λ . Mutual information is $I(\lambda) = \log 2 - H_b(\sigma(\lambda))$.

Empirical MI estimates use fixed seed 0 and share the same λ grid as the closed-form sweep in `output/data/parameter_sweep.csv`.

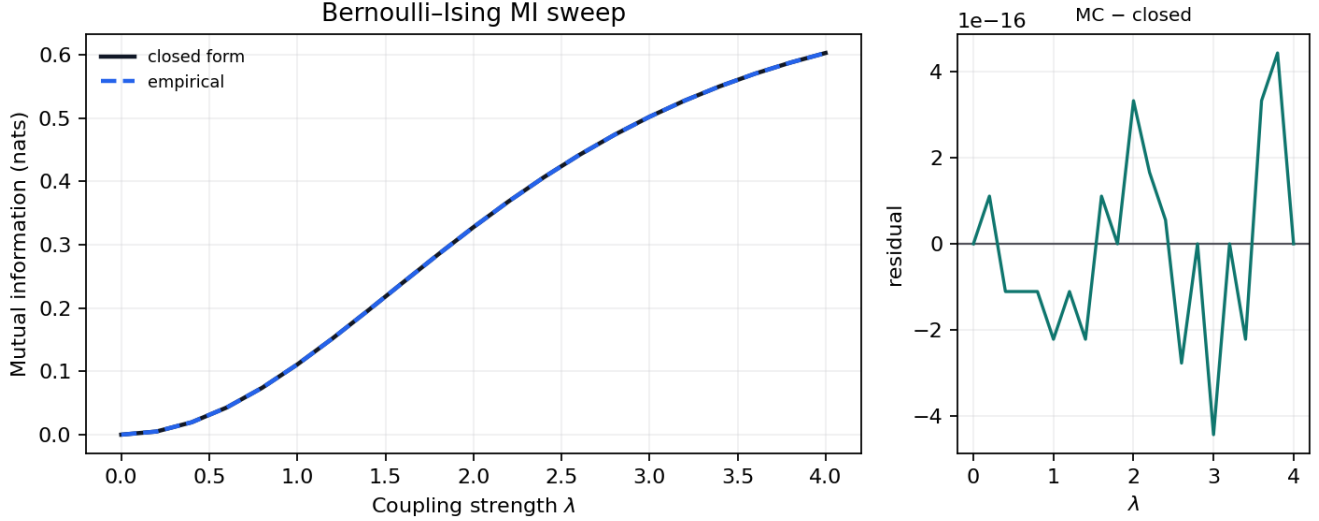


Figure 6: Two-panel plot of mutual information versus coupling strength λ for the symmetric Bernoulli-Ising toy. Left panel shows closed-form curve (solid dark line) and Monte Carlo estimates (dashed blue line) with λ on the x-axis and mutual information in nats on the y-axis. Right panel shows Monte Carlo minus closed-form residuals versus λ with a zero reference line.

Figure 1. Closed-form and Monte Carlo estimates of pairwise mutual information in the symmetric Bernoulli-Ising toy; saturation approaches $\log 2$ nats.

10 Free energy decomposition

Free energy against the entangled prior is evaluated along the same λ grid used for the MI sweep. The curve highlights how deformation strength trades off prior alignment and coupling energy in the analytical toy. Saturation MI reference: 0.6931 nats.

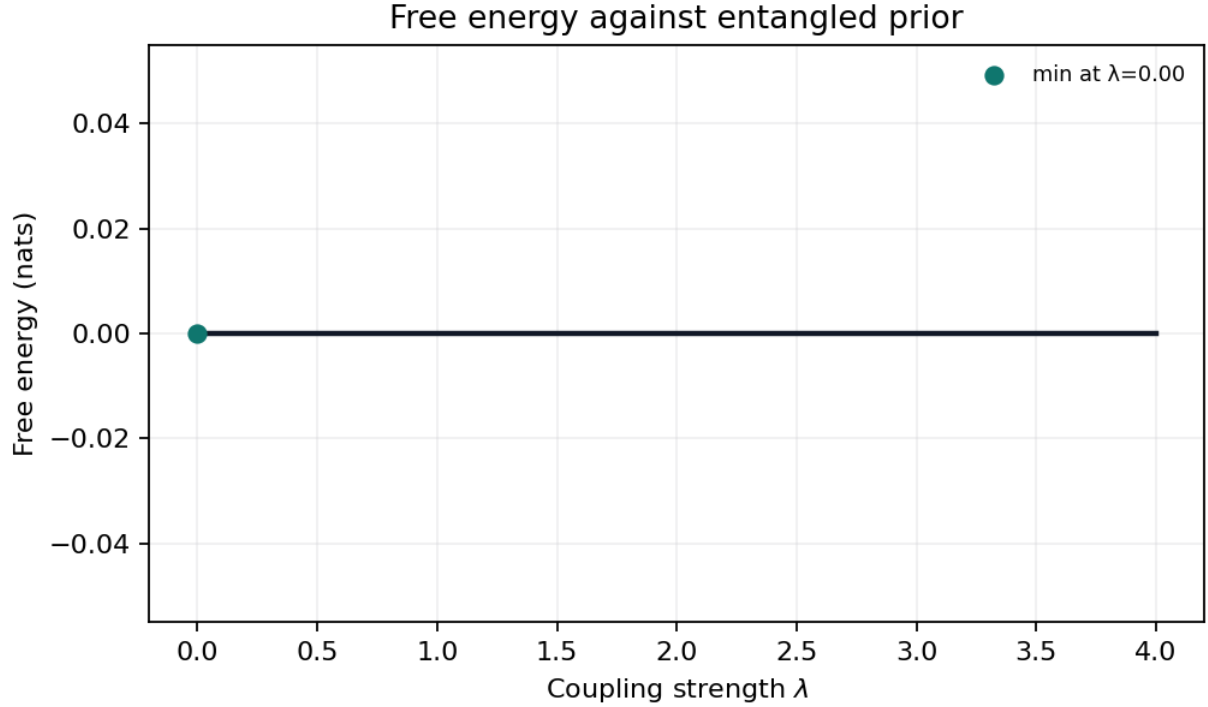


Figure 7: Line plot of free energy in nats versus coupling strength λ for the entangled posterior relative to a mean-field prior. A single dark curve traces free energy across the sweep; the minimum is marked with a teal scatter point and labeled with the argmin λ value.

Figure 2. Free energy of the entangled posterior relative to the mean-field prior across the hyperparameter sweep (grid points 21).

11 T-maze sophisticated inference rollout

The pymdp harness rolls out a T-maze sophisticated-inference agent with planning horizon 2. Summary metrics land in `output/data/si_tmaze_summary.json`.

Steps recorded: 2. Mean belief entropy: 0.3251.

Rollout trace: `output/data/si_tmaze_trace.json`. JSONL run log: `output/logs/pymdp_runs.jsonl`.

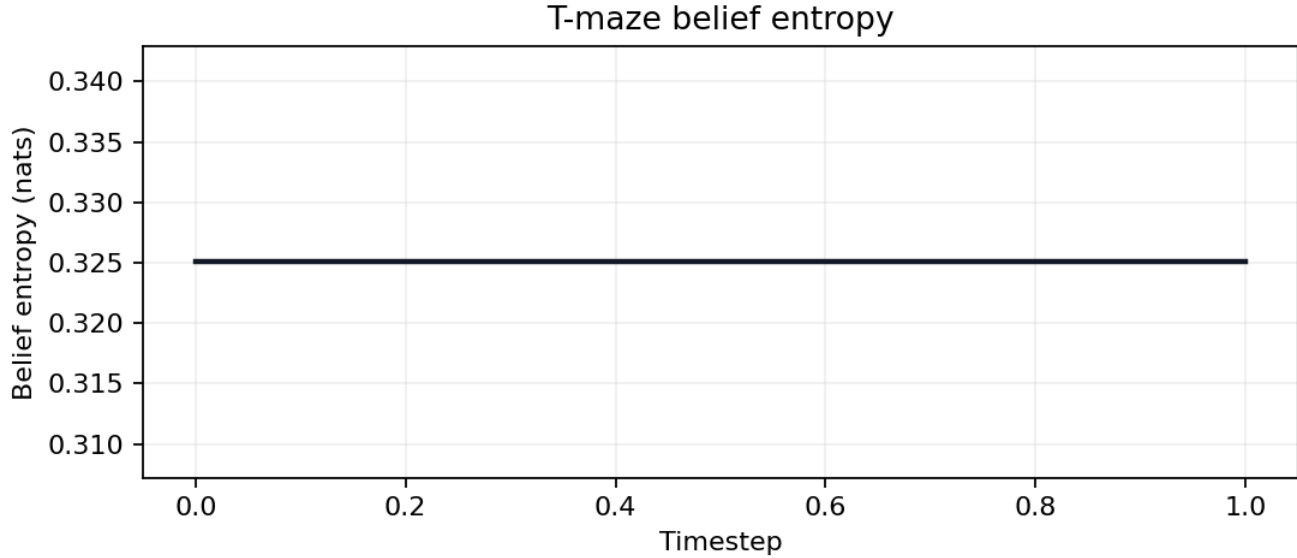


Figure 8: Line plot of belief entropy in nats versus timestep for the pymdp T-maze rollout. Entropy ranges from 0.3251 to 0.3251 nats across 2 steps in `state_inference` mode.

Figure 3a (results). Belief entropy over time for the T-maze rollout (mean 0.3251 nats).

Figure 3b (results). Observation and action traces for the T-maze rollout (action diversity 2).

Figure 3c (results). Discrete action index over time for the pymdp T-maze rollout (policy length 2).

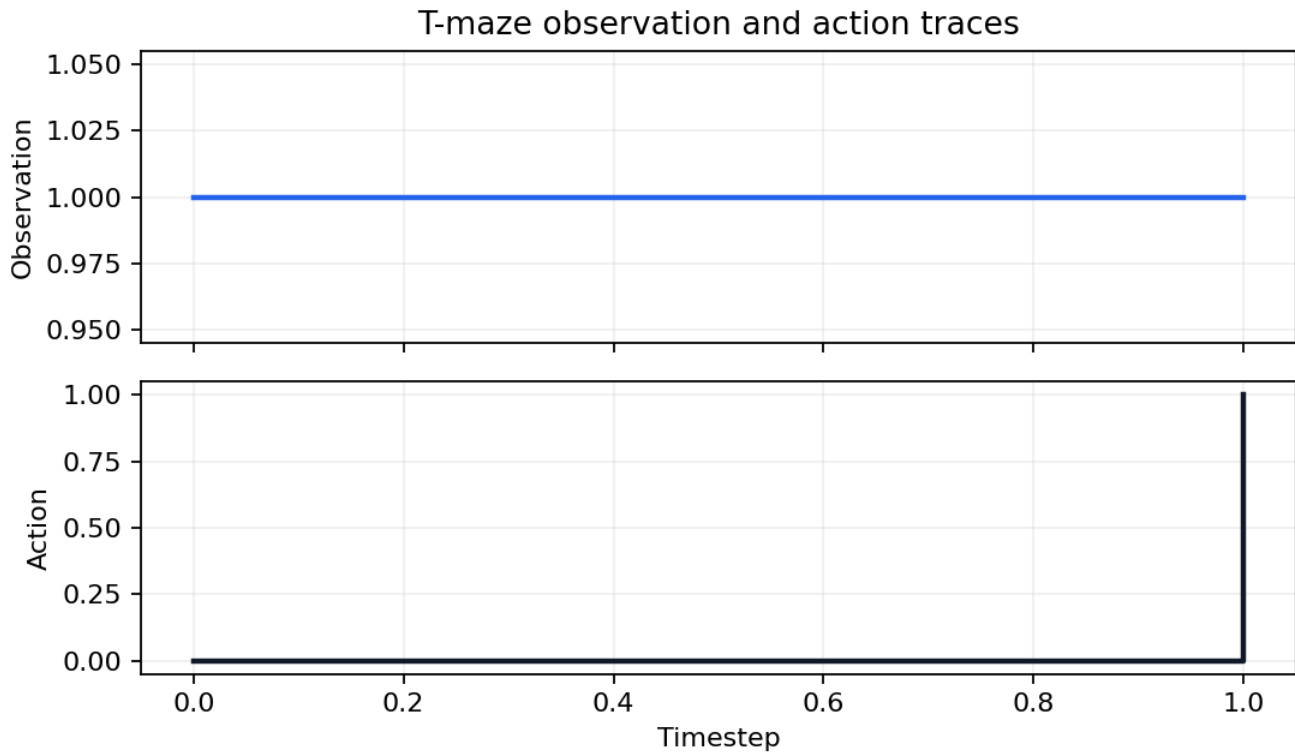


Figure 9: Dual-panel plot of observation index and action index versus timestep for the pymdp T-maze rollout. The upper panel shows discrete observations; the lower panel shows actions. Goal reached flag is 1.

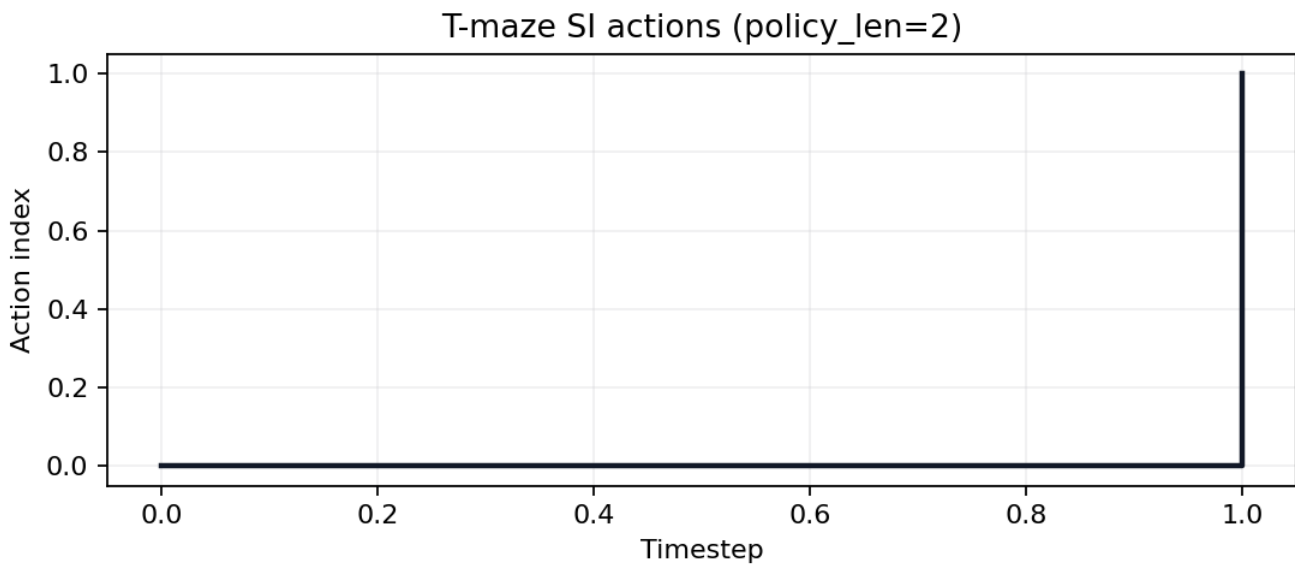


Figure 10: Step plot of discrete action index versus timestep for the pymdp sophisticated-inference T-maze rollout. Actions change at each timestep with light fill under the step trace; policy depth is 2 steps.

12 Invariant gate summary

The analytical invariant registry (`src/invariants.py`) runs before PDF rendering. On a clean checkout **6 / 6** checks pass; failures block publication artifacts.

Report path: `output/reports/invariants.json`.

13 Limitations and extensions

Limitations. The Bernoulli–Ising toy and T-maze harness are pedagogical — they validate pipeline wiring, not empirical claims about biological agents. Default pymdp mode is `state_inference` with planning horizon 2; full policy inference remains an opt-in extension.

Sheaf audit. The coverage heatmap and appendix nine-track proof make binding state auditable: 42 present / 42 bound / 0 missing cells across 16 manifest rows. Strict compose validation fails on gray cells so downstream PDF rendering inherits an honest matrix.

Extensions. Optional animation and graph-world SI tracks (`tracks.yaml extension_tracks`) add artifacts without expanding the default IMRAD matrix. Promote a private project by copying the sheaf manifest pattern under `man uscript/sections/imrad/`.

Measured pymdp rollout (`state_inference`, config hash `ea4b126a9c22f22f`): mean belief entropy 0.3251 nats over 2 steps; goal reached flag 1; action diversity 2.

Analytical sweep residual RMSE 0 nats (max residual 0). Coverage audit: 42 present / 42 bound / 0 missing cells on the IMRAD matrix.

13.0.1 Ontology bindings

- `terms` → `{‘pedagogical_scope’: {‘label’: ‘Pedagogical scope’, ‘description’: ‘Analytical and pymdp artifacts validate pipeline wiring, not biological claims.’}, ‘state_inference_mode’: {‘label’: ‘State inference harness’, ‘description’: ‘Default T-maze mode runs infer_states with declared policy_len; full infer_policies is opt-in.’}, ‘coverage_semantics’: {‘label’: ‘Coverage matrix semantics’, ‘description’: ‘Black cells are bound and present; white unbound; gray bound but missing.’}}`

14 Full sheaf track coverage (proof)

This section is the **composability proof**: every registered sheaf track is bound and rendered in one flat manuscript section.

For each track $t \in \mathcal{T}$, the manifest binds a fragment path $f(t)$ and the composer emits `<!-- sheaf-track:t -->` before the rendered body.

$$|\mathcal{T}_{\text{full}}| = 9$$

Analytical artifacts: `output/data/parameter_sweep.csv` and `output/reports/invariants.json` from the Bernoulli/Ising sweep track.

pymdp harness summary: `output/data/si_tmaze_summary.json` (mean belief entropy, action trace). Full log: `output/logs/pymdp_runs.jsonl`.

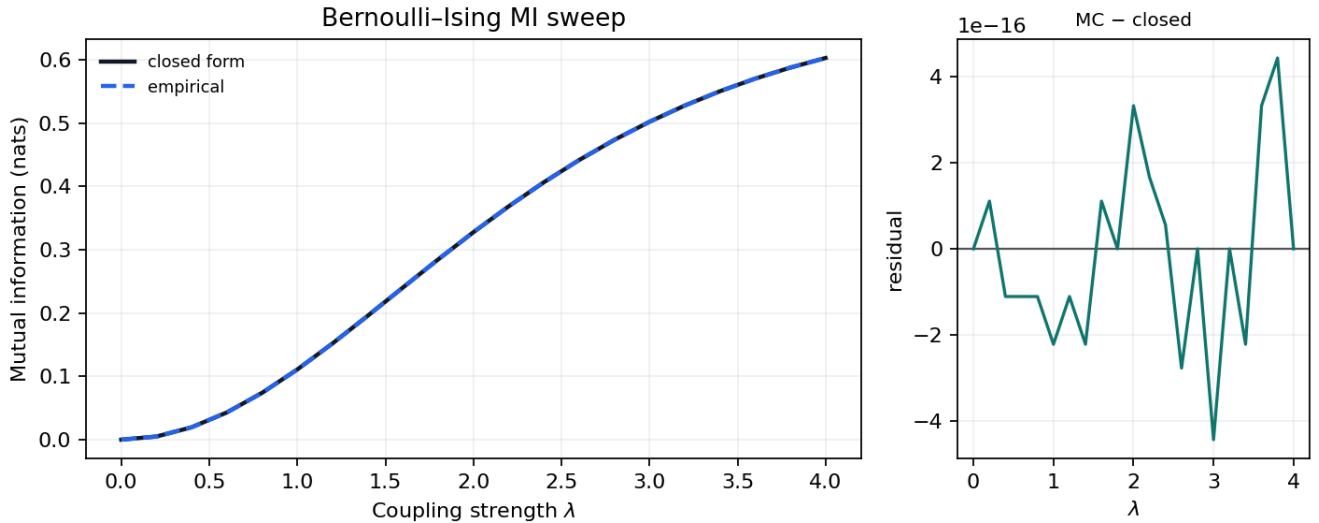


Figure 11: Two-panel plot of mutual information versus coupling strength λ for the symmetric Bernoulli-Ising toy. Left panel shows closed-form curve (solid dark line) and Monte Carlo estimates (dashed blue line) with λ on the x-axis and mutual information in nats on the y-axis. Right panel shows Monte Carlo minus closed-form residuals versus λ with a zero reference line.

Closed-form and Monte Carlo estimates of pairwise mutual information in the symmetric Bernoulli-Ising toy; saturation approaches $\log 2$ nats.

Discrete action index over time for the pymdp T-maze rollout (policy length 2).

Figure 4. Sheaf track coverage for 16 manifest rows and 10 fragment tracks (42 present / 42 bound / 0 missing).

Lean modules under `lean/TemplateActiveInference/` declare horizon and coupling witnesses. Build with `lake build` in `lean/`; some lemmas remain `sorry` by design in this exemplar.

GNN declarations: `gnn/bernoulli_toy.gnn.md` and `gnn/si_tmaze.gnn.md`. Ontology concordance is validated against Active Inference Ontology bindings.

14.0.1 Ontology bindings

- `belief_entropy` → **BeliefEntropy**
- `expected_free_energy` → **ExpectedFreeEnergy**
- `location` → **HiddenState**
- `observation` → **ObservationLikelihood**
- `policy` → **PolicyPosterior**
- `sheaf_track` → **SheafFragment**

Animation track (optional registry flag): static frame from the SI action trace figure.

Future work may replace this with a true frame sequence under `../figures/animation/`.

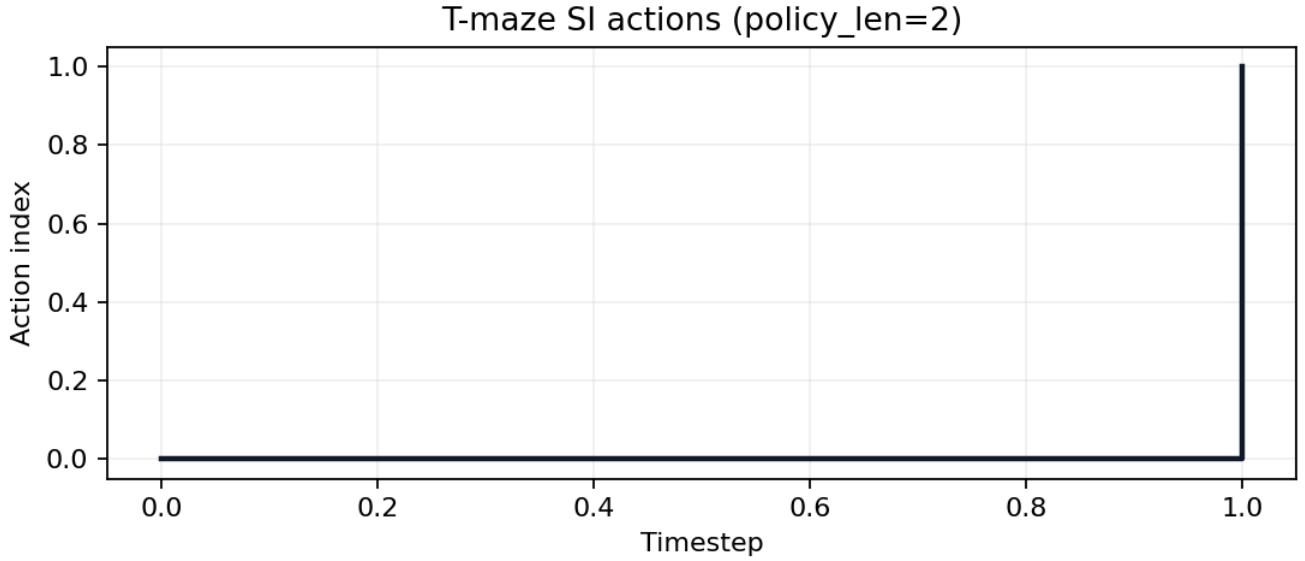


Figure 12: Step plot of discrete action index versus timestep for the pymdp sophisticated-inference T-maze rollout. Actions change at each timestep with light fill under the step trace; policy depth is 2 steps.

15 Conclusion

This exemplar shows how pipeline tracks, sheaf fragment registries, and IMRAD manuscript sections fit together: analytical oracles and pymdp rollouts produce measured artifacts; composition binds 12 flat sections from 10 fragment tracks; coverage JSON and the first-page heatmap report 42 present / 42 bound / 0 missing cells.

Strict validation (`compose_manuscript.py --validate-only --strict`) fails on gray matrix cells, keeping the appendix full-track proof and Results sections trustworthy for downstream PDF rendering. The T-maze harness runs in `state_inference` mode with config hash `ea4b126a9c22f22f`; sweep RMSE 0 nats bounds analytical-empirical agreement on the toy.

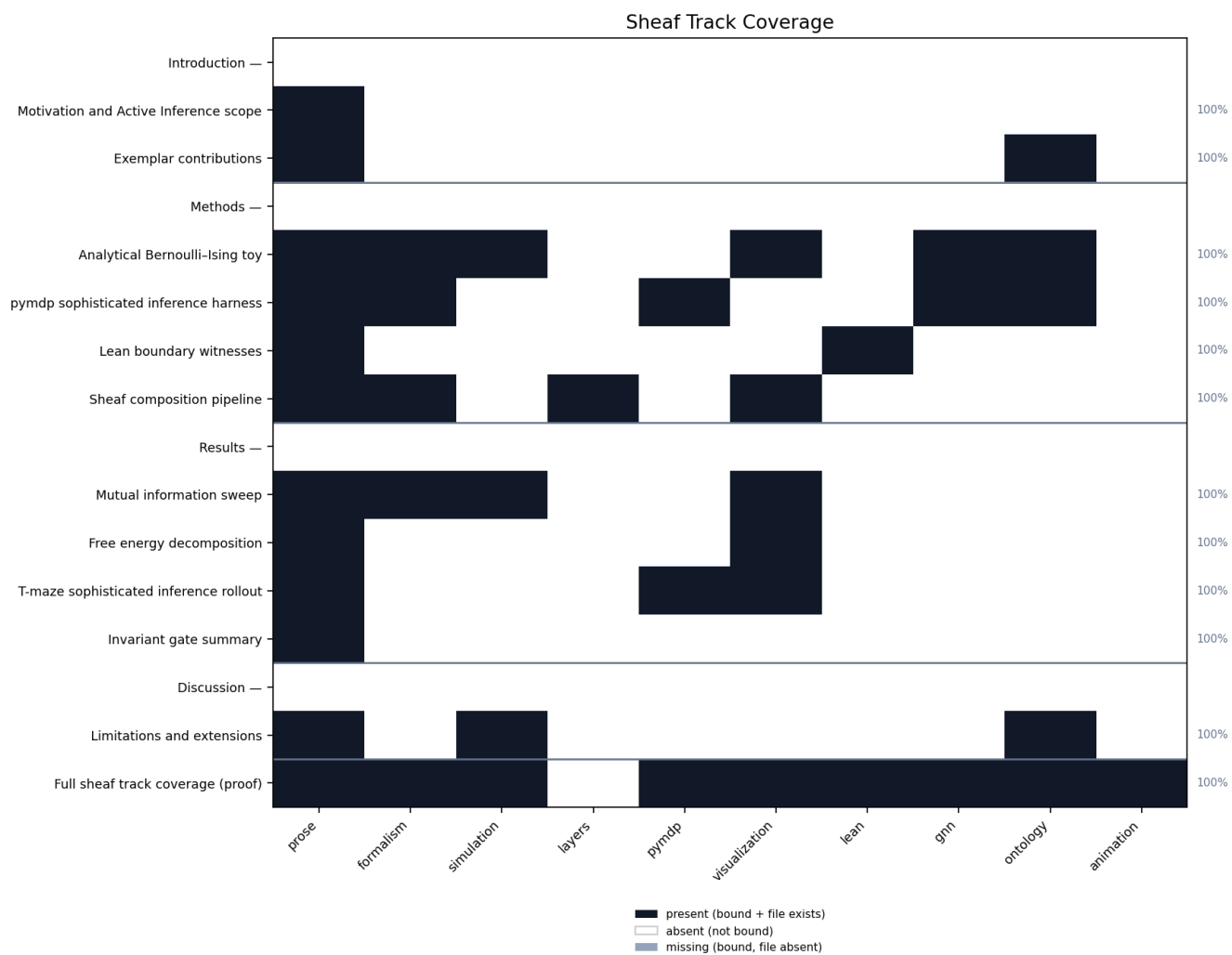


Figure 13: Heatmap matrix of IMRAD manuscript rows versus nine sheaf fragment track columns. Black cells mean the track is bound and the fragment file exists; white cells mean the track is not bound; gray cells mean bound but missing. Rows are grouped by IMRAD block with indented subsection labels; column headers list track ids.

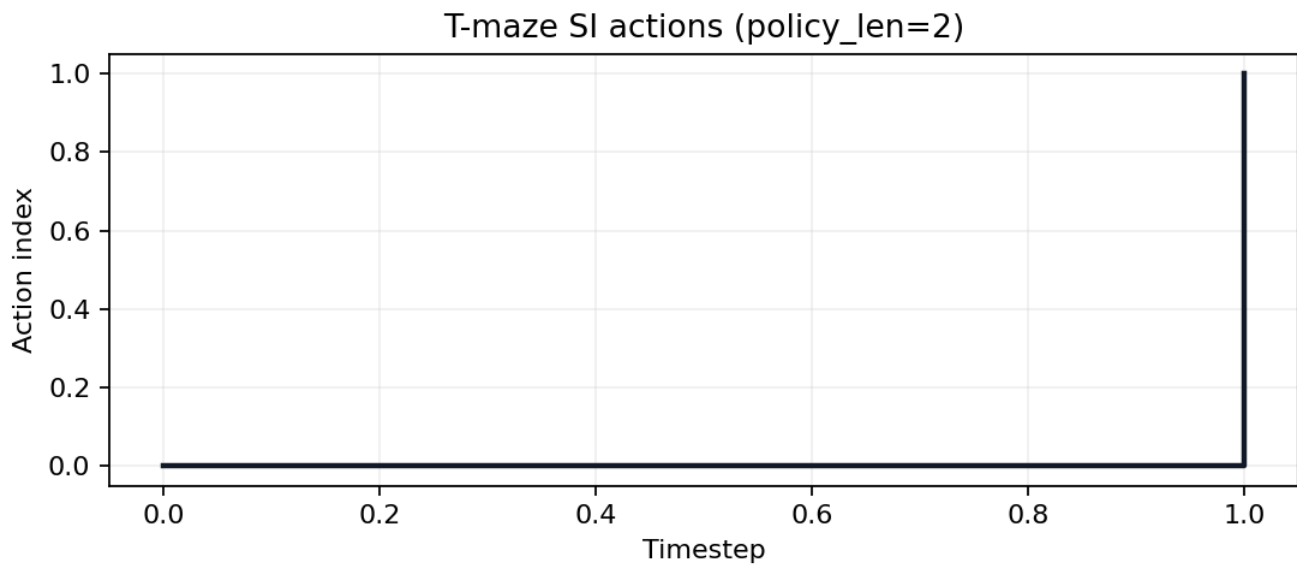


Figure 14: SI action trace frame

16 References

See `manuscript/references.bib` for bibliography entries cited in the composed sections.

References

END OF TRANSMISSION

Release: v0.1.0 · DOI pending · SHA-256 fddea89a41b8... · pairing pending

Figure 15: Integrity QR strip

Prior: *No prior releases.*